DRAFT 3

The minigames

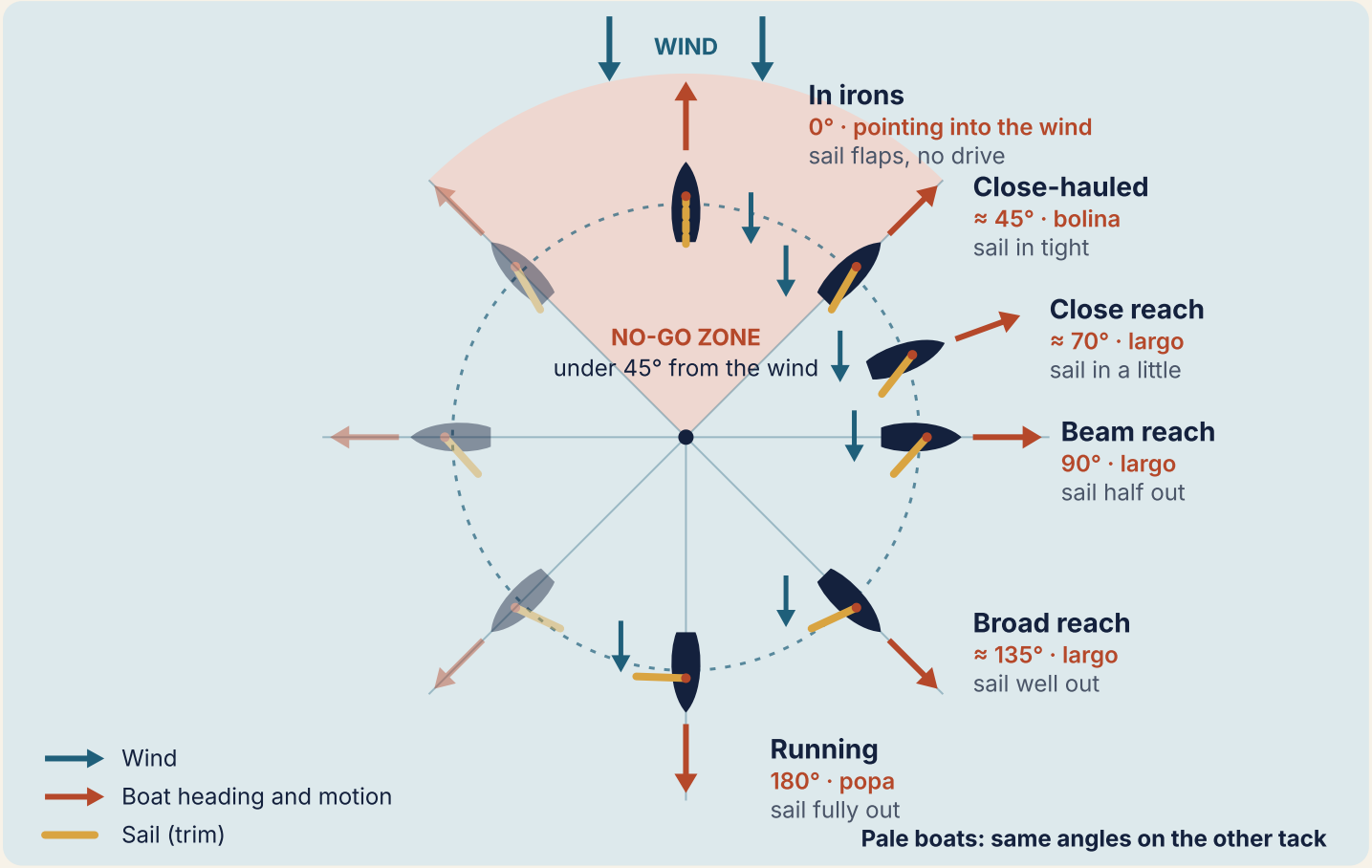
Eight minigames on the Tagus and Portugal's traditional boats. For each one: where it takes place, what you do, and the history behind it. Rules and history are on pages 4 to 7; pages 2 and 3 explain the sailing.

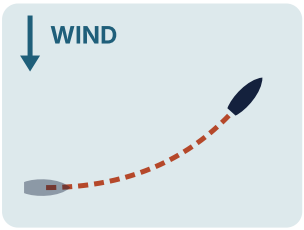
	MINIGAME	WHERE	WHAT YOU DO	WHY IT IS HERE
1	**Downwind with the Falua**	Tagus, off Lisbon	Reach a mark 800 m dead downwind. The wind is set along the open water, so you sail with it behind you.	Fixed-route passenger boat: every trip meant sailing with the wind and against it.
2	**Upwind with the Falua**	Tagus, off Lisbon	Reach a mark 800 m dead upwind, so you have to tack.	Going upriver against the wind meant tacking, or hauling from the bank.
3	**Passengers with the Catraio**	Tagus	Carry passengers from Lisbon to Almada.	Ferrymen carried people between Lisbon and the south bank; 2,000+ boats by the 1850s.
4	**Whale watching with the Baleeiro**	Azores, from Horta	Come up behind a sperm whale without scaring it. The ending says you watched it, not harpooned it.	Azorean whaling ended in 1984; the game turns the hunt into watching.
5	**Seaweed with the Moliceiro**	Ria de Aveiro, off Torreira	Gather six floating mats of seaweed.	The boat of the Aveiro lagoon, built to harvest moliço for fertiliser.
6	**Cargo with the Fragata**	Tagus	The busy Almada → Lisbon crossing, with other boats, gusts and shifting wind.	The cargo workhorse of the Tagus, in a river with about 1,490 craft in 1552.
7	**Fishing with the Muleta**	Tagus, off Seixal	Drift over the shoals with the net out.	Estuary trawler of Seixal and Barreiro; its net was banned, yet tolerated.
8	**Supplies with the Canoa**	Tagus, off the Bugio	Out to the Muleta at anchor and back.	Sardine-trade boat that left the Tagus with salt and worked at sea.

DRAFT 3

Points of sail

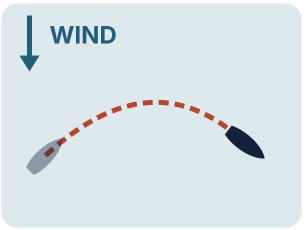
The wind decides where a boat can go. Each boat starts at the centre and sails outward on its heading, with the wind blowing from the top. Angles are approximate: traditional rigs point less close to the wind than modern boats [verify for each boat].





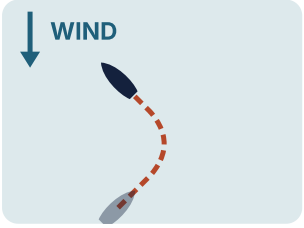
Head up · orçar

Steer toward the wind. The boat points higher and sails closer to it, so sheet in (caçar). Overdo it and it stalls in irons.



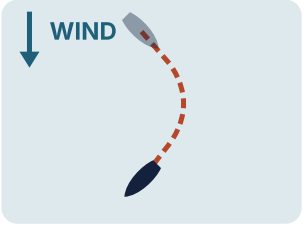
Bear away · arribar

Steer away from the wind. The boat turns downwind and gains speed, so ease the sail (folgar). It is also the way out of irons.



Tack · virar de bordo

Turn through the wind to switch sides while going upwind. The boat slows in the turn, so turn briskly and re-trim.



Gybe (jibe) · cambar

Turn with the wind astern so the sail swings across with force. Steer smoothly. On a lateen rig the yard passes the mast.

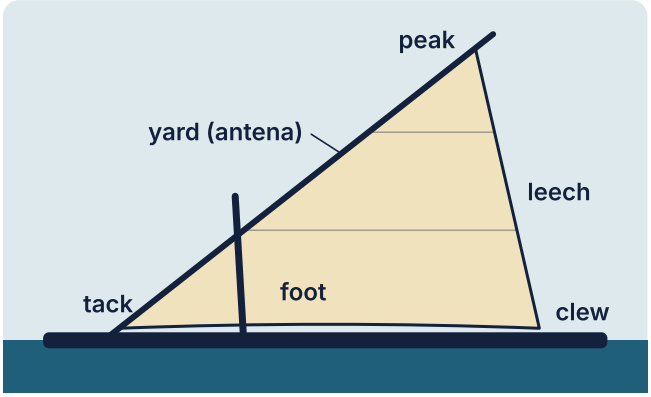
Trim. Caçar pulls the sail in towards the centreline; folgar lets it out. The closer to the wind, the tighter the sail. The two falua minigames practise it: downwind and upwind.

DRAFT 3

The four sails

Four sail shapes cover the boats in the game. Each one changes how a boat turns, what it does best and how you handle it. Drawings are schematic; boat lists and local names are to be confirmed with the Museu de Marinha.

Lateen vela latina



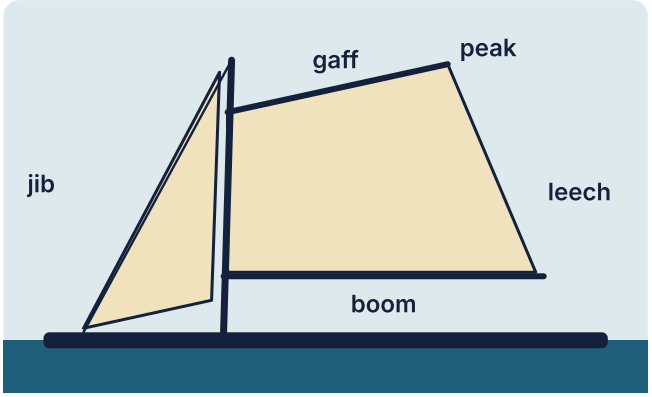
Shape. A triangle hung from a long yard that slopes up across a short mast.

Boats. Falua, muleta, canoa da picada.

Handling. To change sides with the wind astern, the yard is passed to the other side of the mast. The canoa da picada did it in a light breeze from astern, and it took great skill.

In the game: Passing the yard (cambar), trimming with caçar and folgar.

Gaff · carangueja vela de carangueja



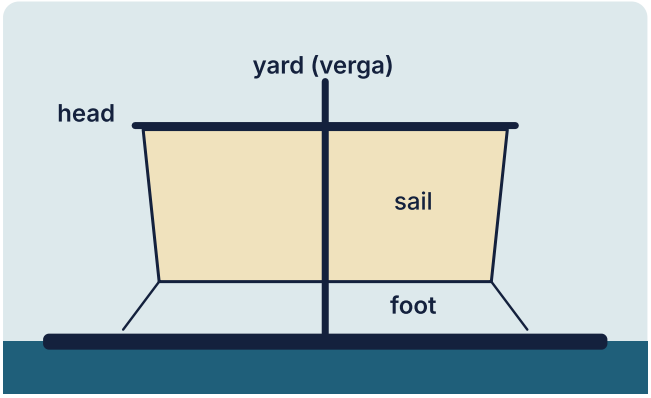
Shape. A four-sided sail between a gaff above and a boom below, set behind the mast, with a jib ahead of it.

Boats. Fragata. Canoa cacilheira.

Handling. The whole sail stays on one side of the mast, so there is no yard to pass. The boom swings across in a gybe. With no oars, a fragata in a calm was towed by a small bote.

In the game: Gaff handling, docking in traffic, gusts.

Square vela redonda



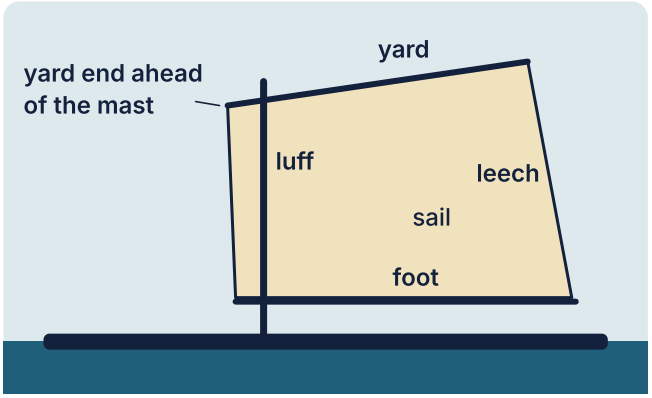
Shape. A rectangle hung from a yard that crosses the mast.

Boats. Varino, in one source. Muleta crossing the river: the 1572 rules ask for a round sail and a foremast.

Handling. Drives the boat best with the wind behind. It is a poor choice for sailing upwind.

In the game: Running and broad reach. [Boats and names to confirm.]

Lug · catita vela de catita [to confirm]



Shape. A four-sided sail on a yard that slopes across the mast, with the head reaching a little ahead of it.

Boats. Catraio. Moliceiro: about 24 m² of sail.

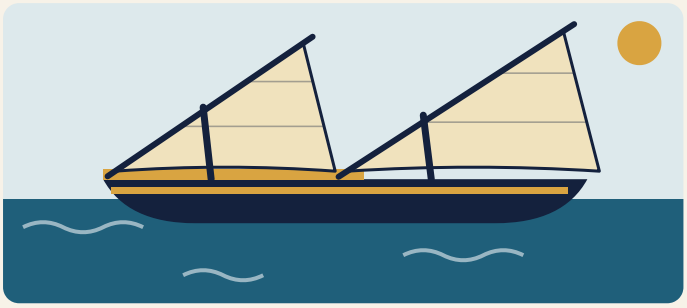
Handling. A lot of sail on a light boat. On the moliceiro, leeboards limit sideways slip in about a metre of water; when the wind drops, the crew poles or tows. [Handling to confirm.]

In the game: Catraio: crossing the current. Moliceiro: leeboards, pole or tow.

The minigames 1 · 2

DRAFT 3

1 Downwind with the Falua



WHERE · TAGUS, OFF LISBON

The game. Reach a mark 800 m dead downwind. The wind is set along the open water, so you sail with it behind you.

RULES

Goal: reach the mark, 800 m dead downwind.

Skill: running and broad reach. Ease the sail (folgar) and avoid an accidental gybe (cambar).

Score: time to the mark [proposal].

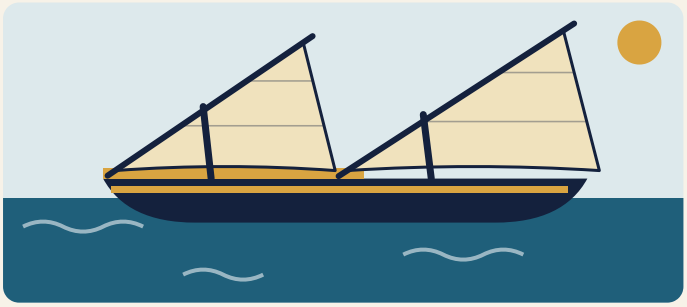
History. The falua carried passengers and fresh produce on a regular, fixed route, up to about 50 km, so a trip meant sailing both with the wind and against it. It had two lateen sails (later often one), a forward deck sheltering passengers from cold, rain and heat, and slender lines, which made it fast for its size. It measured 14 to 16 m with a crew of two or three, between the bote and the fragata in size.

Sources: Revista da Armada no. 470 (2013), citing M. Leitão (2002) and E. Carrasco (1997); jmbd1945 blog.

[Game screenshot]

1 · Downwind with the Falua

2 Upwind with the Falua



WHERE · TAGUS, OFF LISBON

The game. Reach a mark 800 m dead upwind, so you have to tack.

RULES

Goal: reach the mark, 800 m dead upwind.

Skill: close-hauled (bolina) and tacking (virar de bordo). Turn briskly or the boat stalls in irons.

Score: time to the mark [proposal].

History. Going up the Tagus against the wind meant working to windward, tacking from side to side. Where the current was strong, men on land hauled the boats with a rope or pole from towpaths (sirgadouros). Upriver trade ran mainly from April to October, and in the mid-19th century the trip from Santarém to Barquinha took about ten hours.

Sources: F. Freire, "Os cais na Barquinha"; J. Candeias da Silva, "O antigo porto fluvial de Abrantes".

[Game screenshot]

2 · Upwind with the Falua

The minigames 3 · 4

3 Passengers with the Catraio



WHERE · TAGUS

The game. Carry passengers from Lisbon to Almada.

RULES

Goal: take the passengers across the river from Lisbon to Almada.

Skill: crossing the current and coming alongside the quay (approach and dock).

Score: passengers delivered in the time [proposal].

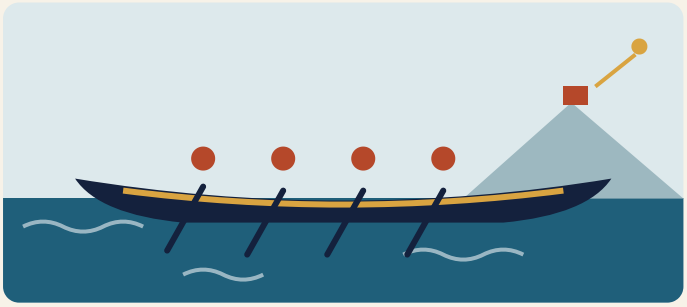
History. The catraios were the small boats of the Tagus ferrymen, the catraeiros, who carried people between Lisbon and the south bank. By the mid-19th century more than 2,000 boats worked between Lisbon and Almada. Steam arrived in the 1860s, yet the traditional boats long resisted. In the mid-1920s boatmen at Porto Brandão founded the Cooperativa de Catraeiros do Porto de Lisboa. The 1966 bridge ended the vehicle ferries. [Catraio size and rig: to confirm.]

Sources: Observador (J. P. Pincha, 2016); Almada Virtual Museum (2016).

[Game screenshot]

3 · Passengers with the Catraio

4 Whale watching with the Baleeiro



WHERE · AZORES, FROM HORTA

The game. Come up behind a sperm whale without scaring it. The ending says you watched it, not harpooned it.

RULES

Goal: close in behind the whale and stay with it.

Avoid: coming in fast, from the front or too close, which scares the whale [limits to confirm].

Ending: you watched the whale; there is no hunt.

History. Azorean whaling developed in the second half of the 19th century. The bote baleeiro, an open boat with a crew of seven, went out under sail or oars and struck with hand-thrown harpoons. On land a lookout, the vigia, watched for the blows and fired a rocket to launch the boats. The hunt ended in 1984 under the international ban. Regattas keep the boats alive, and in July 2026 a consultation opened on listing the bote's construction and handling as national intangible heritage.

Sources: Açoriano Oriental (6 Jul 2026); blog "Baleias e Baleeiros" (2011).

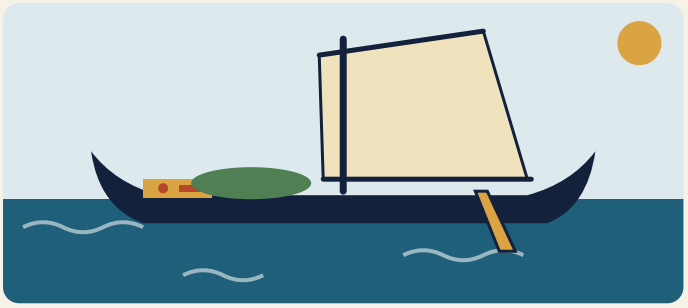
[Game screenshot]

4 · Whale watching with the Baleeiro

DRAFT 3

The minigames 5 · 6

5 Seaweed with the Moliceiro



WHERE · RIA DE AVEIRO, OFF TORREIRA

The game. Gather six floating mats of seaweed.

RULES

Goal: gather all six floating mats.

Skill: sailing in about a metre of water with leeboards; keeping the load balanced.

Score: mats gathered and time [proposal].

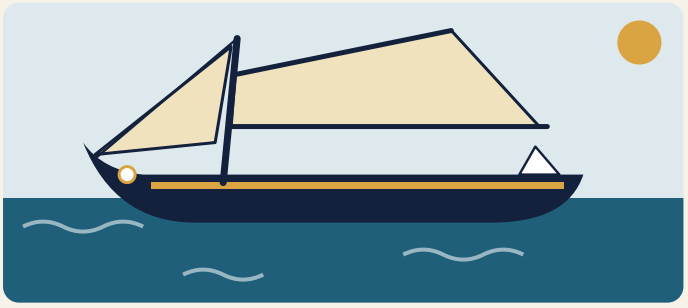
History. The moliceiro was built to gather moliço, the aquatic plants of the Ria de Aveiro (the game calls it seaweed), the main fertiliser of Aveiro's farmland. It is about 15 m long and 2.5 m wide, with a flat bottom that works in about a metre of water, and carries roughly 5 tonnes. A lug sail of about 24 m² and two leeboards drive it; crews could also pole or tow it. Each boat has four painted panels. Harvesting ended in the 1980s and 1990s, and tourism saved the boat.

Sources: INPCI, "Barco Moliceiro" (INPCI_SU_2022_003); Wikipedia PT, "Moliceiro".

[Game screenshot]

5 · Seaweed with the Moliceiro

6 Cargo with the Fragata



WHERE · TAGUS

The game. The busy Almada → Lisbon crossing, with other boats, gusts and shifting wind.

RULES

Goal: bring the cargo across from Almada to Lisbon.

Avoid: collisions with the other boats; gusts and wind shifts call for quick trim.

Score: cargo delivered, collisions [proposal].

History. The fragata was the cargo workhorse of the Tagus: 20 to 25 m long, with one mast, a gaff mainsail and a jib, a crew of three to five and no oars. It unloaded ships at anchor in Lisbon and served the quays on both banks. In calms a small bote towed it. In 1552 about 1,490 river craft worked the Tagus, and by the mid-19th century more than 2,000 boats worked between Lisbon and Almada. Steamers, railway and bridges ended the trade; the 1966 bridge finished it.

Sources: L. Bayó Veiga, getLISBON (2022); Almada Virtual Museum (2016); +Lisboa (2021).

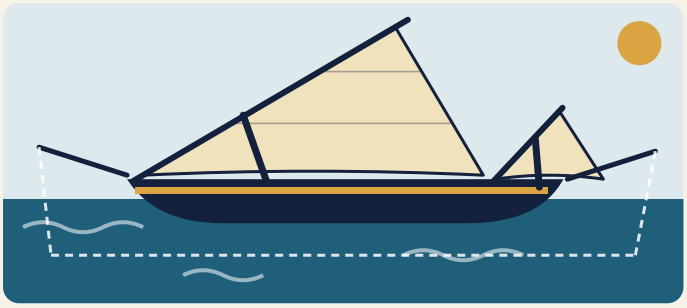
[Game screenshot]

6 · Cargo with the Fragata

DRAFT 3

The minigames 7 · 8

7 Fishing with the Muleta



WHERE · TAGUS, OFF SEIXAL

The game. Drift over the shoals with the net out.

RULES

Goal: drift over the shoals with the net out and haul in a catch.

Skill: lying across wind and current, drift, adjusting the many sails.

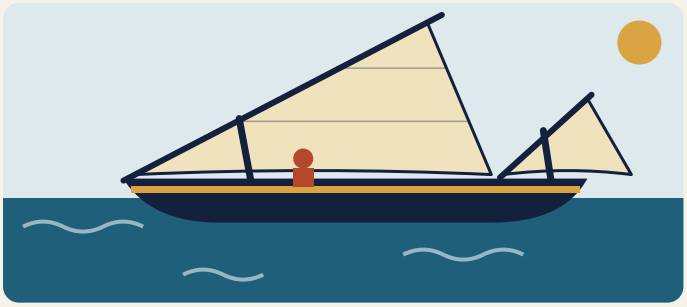
Score: fish caught [proposal].

History. The muleta, armed from Seixal and Barreiro, fished with the tartaranha, a trawl dragged across the beam while the boat lay across wind and current, mainly for flatfish. The net was banned by a royal letter in 1470 and again in 1646, yet it was tolerated when fishermen petitioned; in 1647 it was allowed outside the bar, with March to May closed for spawning. Descriptions give it a sail plan of many sails, with long poles (batelós) to sheet them and tie the net warps. The sources disagree on when it ended.

Sources: F. G. Pedrosa, "A Muleta e a Tartaranha"; Museu de Marinha (MM.05440); Baldaque da Silva (1895).

[Game screenshot]
7 · Fishing with the Muleta

8 Supplies with the Canoa



WHERE · TAGUS, OFF THE BUGIO

The game. Out to the Muleta at anchor and back.

RULES

Goal: sail out to the muleta at anchor and return with the supplies.

Skill: reefing in gusts and passing the yard (cambar).

Score: time for the round trip [proposal].

History. The canoas da picada were sailing boats of the fish trade. They bought sardines from fishing boats outside the harbour and carried them to Lisbon; the fish were salted on board, which is how "picada" came to complete the boat's name. They left the Tagus loaded with salt and worked the Sado and the south coast. They were 12 to 18 m long, with a crew of seven and a large lateen sail, a small mizzen and a storm sail. The only surviving original, Fatinitza (1903), is at the Museu de Marinha.

Sources: V. Moura Vicente, mourav.com (no sources cited) [cross-check with the museum]; jmbd1945 blog.

[Game screenshot]
8 · Supplies with the Canoa

DRAFT 3

A river with history

1785
João de Souza's album of all the boats of the Tagus is engraved, with a plate of fragatas. [One source gives 1875: verify.]

1807–1810
As the French invade, the royal court sails from the Tagus for Brazil (29 Nov 1807). The Lines of Torres Vedras defend Lisbon.

1860s
Steam arrives: the Empresa de Vapores Lisbonenses takes over the Cacilhas crossing.

1966–1975
The bridge ends the vehicle ferries (1966). The crossings are nationalised as Transtejo (1975).

1984
Azorean whaling ends under the international ban.

What did the river's boats do during the French invasions (1807–1810)? The sources read so far do not say. Research in progress with [Museu de Marinha] and [Ecomuseu do Seixal].

Further reading

Luís Ribeiro, Histórias do Tejo (Esfera dos Livros, 2012), the inspiration for several minigames.

António Nabais, Barcos (Câmara Municipal do Seixal, 1982); Navegando no Tejo (CCRLVT, 1995).

M. Leitão, Barcos do Tejo (Museu de Marinha, 2002); E. Carrasco, Barcos do Tejo (Inapa, 1997).

F. G. Pedrosa, "A Muleta e a Tartaranha (séculos XV–XX)".

A. A. Baldaque da Silva, Estado actual das pescas em Portugal (1891), and "A muleta de pesca", Arte Portugueza (1895).

Credits [confirm before print]

Game: "A sailing simulator for the revival of the Tagus River nautical heritage" (DAACH 2025).

[Partner and funder logos]

[QR]

Play the web version

[short URL]